

AP157 Output 11 - April 27, 2026

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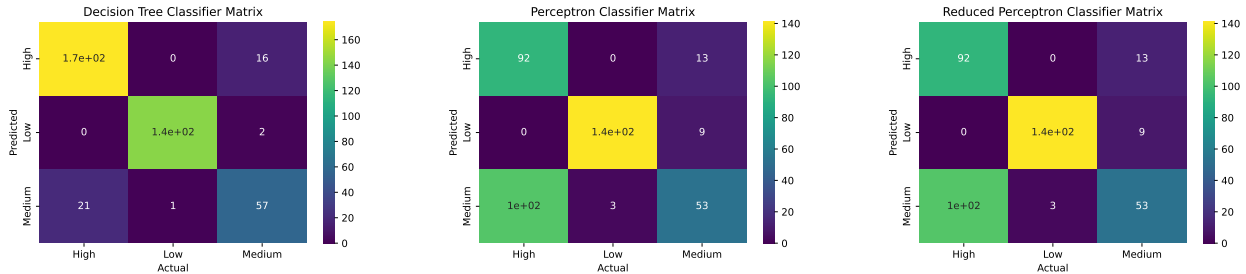


Figure 1: **Classifiers** We assess the prediction of our classifier models Decision Trees and Perceptron for total money and traded, units traded, largest single transaction, smallest single transaction, percent change calculation, days since last login and trade, as well as met realized losses and gains. For reduced perceptron, we eliminated login details as well as extreme trading and percent change data in place of income level.

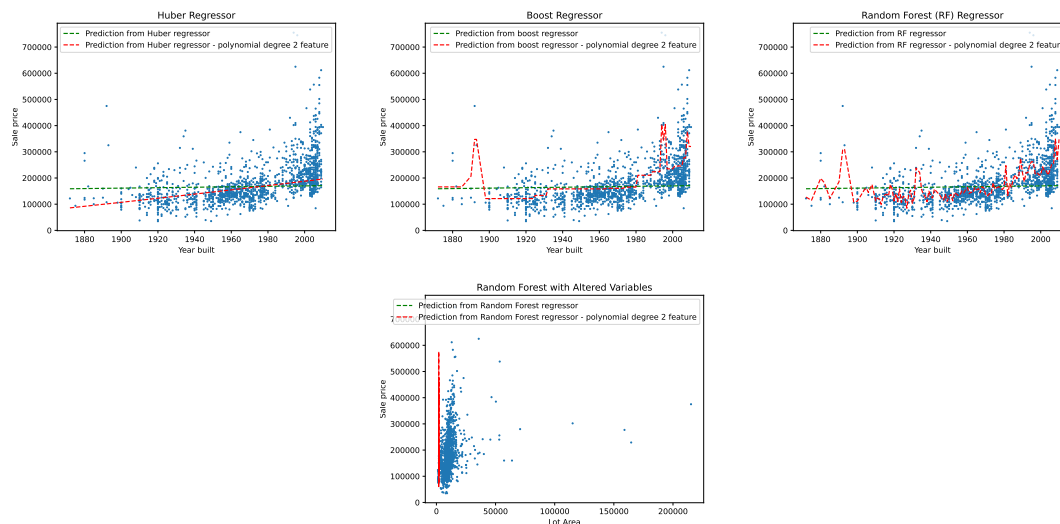


Figure 2: **Regression** Plots showing the regression between year of house's construction, and sale price under Huber (Linear), Boost, Random Forest. Meanwhile, Random Forest is repeated in lot area vs. year

Classifier Matrices

Look here that Decision Trees (non-reduced) is the most accurate model here with most of predictions in a category correct. In perceptron models, they have a tendency to lag behind predicting high churn rates. Even if we reduce the parameters, the output is the same for the perceptron layer implying that the reduced parameters are at least enough for prediction.

Regression Graphs

Huber Regressor performed the best in Year Built vs. Sale Price, albeit with inability to capture non-linearity even in degree 2 or more. Random Forest overestimates the noise of the scatter points by introducing noise while Boost Regression exaggerates the gradient. Despite Random Forest's supposed noise, it is linear in the lot area vs. sale price while missing most of scatter points. Kernel ridge (not shown here) was tried but it hit a divergence.